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Identification of factors associated with foreign substance contamination in restaurant food

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This study identified factors associated with foreign substance contamination in food prepared at restaurants in a district of Seoul. Data from 4,747 registered general restaurants were analyzed using the Hygiene Administration System and the Integrated Food Safety Information System. The dependent variable was the report of foreign substance contamination, and independent variables included owner characteristics (e.g., age group) and restaurant characteristics (e.g., cuisine type). Logistic regression analysis with stepwise selection was performed to identify factors associated with foreign substance contamination. Foreign substances were detected in 343 restaurants (7.2%), with hair was the most frequently detected, followed by pests, metal, plastic, and plastic wraps. The likelihood of contamination was significantly higher in restaurants operated by owners in their 30s, jointly operated by two or more individuals, using third-party delivery platforms, and serving Chinese cuisine. The findings suggest that foreign substance contamination in restaurants is influenced by complex and context-specific conditions, underscoring the need for tailored, risk-based hygiene strategies. However, given that the analysis was restricted to one district in Seoul, caution should be exercised when interpreting the generalizability of the findings.

Keywords Foreign substance contamination, Restaurant hygiene, Delivery platform, Multiple owner

In Korea, the number of foreign substance reports increased almost fivefold from 2,615 in 2016 to 13,963 in 2022¹, highlighting the growing urgency of this public health concern. Foreign substances in food refer to non-edible materials unintentionally mixed into food products, including metal, glass, plastic, insects, and hair, which can pose health risks to consumers^{2,3}. The Codex Alimentarius, jointly developed by the Food and Agriculture Organization (FAO) and the World Health Organization (WHO), classifies such physical hazards alongside biological and chemical hazards as critical food safety risks⁴, and approximately 14% of reported foreign object incidents have resulted in injury or illness^{5–7}. Contamination not only undermines consumer trust but also damages brand reputation and poses a threat to public health. However, most of these studies remain limited to industrial food processing contexts, and few empirical investigations have examined contamination risks in restaurant or delivery-based environments.

Meanwhile, the COVID-19 pandemic has dramatically transformed the dining landscape. The rise of “dark kitchens” or “ghost kitchens”—delivery-only establishments without customer-facing space—has disrupted traditional foodservice models^{8–12}. These kitchens operate exclusively through online platforms, and their cooking environments are not visible to consumers^{13–16}. Often, they lack basic sanitation infrastructure such as proper ventilation, hot water, or adequate space¹⁷. Consumer concerns about cleanliness have increased^{18,19}, raising regulatory challenges as these establishments are not clearly distinguishable from conventional restaurants¹⁷. To address these emerging challenges, the present study extends previous research—largely confined to industrial food processing—by analyzing administrative data from Korea’s Integrated Food Safety Administration System and Hygiene Administration System, covering both restaurants with and without reported contamination incidents. This broader scope allows us to identify structural and operational risk factors in restaurant and delivery-based environments that have been underexplored in prior studies.

Previous research on foreign substance contamination has primarily focused on manufactured and packaged foods. The Rapid Alert System for Food and Feed (RASFF), operated by the European Union, is a platform for swiftly sharing information on food and feed safety hazards. Studies have analyzed RASFF data in Europe

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to identify regional differences in foreign object types and associated food categories²⁰. Others have reported frequent detection of glass, plastic, and animal material in vegetables, grains, and meat products in the UK²¹. Brazilian consumer complaint data also showed contamination cases involving plastic, insects, and bone fragments in dairy and meat products^{22,23}. Additional studies have traced metal fragments in food to specific stages of the production process²⁴, highlighting equipment damage, packaging failure, and poor sanitation as major sources of contamination²⁵. According to a Korean Consumer Agency survey, restaurant or delivery foods accounted for the largest proportion (19.7%) of all foreign object incidents²⁶. Recent findings also suggest that over 10% of food recalls are due to foreign matter, with plastic being the most common contaminant²⁷.

Food hygiene is known to be influenced not only by individual-level factors, but also by a complex interplay of conditions including kitchen environment, hygiene training, food safety knowledge and behavior, hands-on experience, and institutional support. For instance, Augustin et al.²⁸ reported that the physical structure of kitchens influences hygiene practices²⁸. Hann et al.²⁹ demonstrated that the SafeConsume program enhanced food safety knowledge and behavior among adolescents²⁹, while Liu and Lee³⁰ emphasized the role of institutional trust and perceived managerial accountability in shaping hygiene compliance³⁰. From the consumer perspective, Bai et al.³¹ found that food safety perceptions in restaurant settings are shaped by prior knowledge, personal experience, and environmental cues, including staff hygiene and kitchen visibility³¹. Similarly, Kasza et al.³² underscored a gap between food safety awareness and actual consumer behavior, highlighting the need for effective education and institutional reinforcement³².

Prior research indicates that restaurants serving Chinese cuisine in Korea tend to have lower sanitation standards compared to those serving Korean, Japanese, or Western cuisine³³. Ethnic restaurants, often run by minorities, tend to experience more violations due to language barriers, limited hygiene training, and lack of equipment³⁴. To fill this gap, the present study analyzes administrative data from Korea's Integrated Food Safety Administration System, comparing restaurants with and without reported contamination incidents. By investigating how factors such as delivery platform registration affect the likelihood of contamination, the study aims to identify structural risk factors in the evolving foodservice landscape and inform targeted hygiene management strategies.

This study hypothesizes that the characteristics of restaurants (e.g., cuisine type, floor level, and use of delivery platforms) and those of restaurant operators (e.g., age, gender, and number of owners) are associated with the risk of foreign substance contamination.

Methods

Data sources and collection

This study analyzed data from 4,747 general restaurants and 343 foreign substance reports collected between 2021 and 2023 in an anonymous district of Seoul, South Korea. The data were obtained from the Hygiene Administration System (HAS) and the Integrated Food Safety Administration System, both operated by the Ministry of Food and Drug Safety. HAS includes administrative information on restaurant licenses, owner characteristics, floor area, delivery app registration, and location. The Integrated Food Safety Administration System manages consumer- and owner-submitted reports of foreign substances in food.

Restaurants were limited to those classified as general restaurants, excluding café or bakeries due to differences in business type and menu structure. The analysis was geographically limited to a single district in Seoul, and results should therefore be interpreted with caution regarding generalizability. As a secondary analysis using public administrative data, the study received exemption from review by the Institutional Review Board (IRB) of the University of Seoul (UOS-IRB-2024-07-002).

Definition of variables

The dependent variable was whether a foreign substance in food was reported, coded as 1 for reported and 0 for not reported. Independent variables were selected based on theoretical relevance to food hygiene behavior and operational risk, as well as data availability. These were categorized into owner characteristics and restaurant characteristics. From a structural perspective, prior research suggests that kitchen layout and physical space constraints can affect hygiene practices (e.g., workflow, separation of zones)²⁸, while operational complexity—such as reliance on food delivery platforms or the use of ghost kitchens—may introduce time pressure and regulatory blind spots¹⁷. Owner characteristics, including gender, age group, and the number of owners (one vs. two or more), were considered for their relationship to perceived responsibility and hygiene accountability. In particular, longer operator experience has been positively associated with greater awareness and implementation of food safety practices³⁰. Restaurant characteristics included floor area (per 100 m²); use of food delivery platforms (Yes, No); duration of operation measured both at the current location and by the current owner; cuisine type (Chinese, Japanese, Korean, Korean Bunsik, Korean fried chicken, Western, other); floor location (basement, first floor, upper floors); and administrative district (coded anonymously as A–E). Cuisine type and preparation complexity were also included due to their influence on workload and sanitation risks—for example, high-heat, fast-paced cooking environments may increase the likelihood of equipment wear, splattering, and handling errors. From the consumer perspective, hygiene assessments are influenced not only by food appearance but also by staff hygiene, kitchen visibility, and environmental cleanliness³¹. These variables were therefore included as theoretically grounded proxies for structural, behavioral, and perceptual factors associated with food safety risk. These were categorized into owner characteristics and restaurant characteristics.

Owner characteristics included gender, age group, and the number of owners (one vs. two or more). Restaurant characteristics included floor area (per 100 m²); use of food delivery platforms (Yes, No); duration of operation (location, owner); cuisine type (Chinese, Japanese, Korean, Korean Bunsik, Korean fried chicken, Western, other); floor location (basement, first floor, upper floors); and administrative district (coded anonymously as A–E). Use of food delivery platforms was defined as using one or more of the three major domestic platforms,

including Baedal Minjok, Yogiyo, and Coupang Eats. Binary coding (Yes/No) was applied based on official listings. Duration of operation was measured in two distinct ways. The first, operation at the current address (location), was defined as the number of years the restaurant had been operating at the same physical location as of the date of the foreign substance contamination report, regardless of changes in ownership. The second, operation by the current owner (owner), was defined as the number of years the current owner had continuously managed the restaurant as of the same date. Cuisine type was classified based on the Korean Standard Industry Classification (KSIC, 10th revision)³⁵. Korean Bunsik refers to a type of Korean eatery that sells light meals such as gimhap, tteokbokki, instant noodles, and other similar dishes.

Categories comprising less than 4% of the total sample were grouped into “other” for statistical stability. Restaurants with mixed-floor locations (e.g., basement and first floor) were excluded from analysis due to classification ambiguity. Food category refers to the specific type of food in which a foreign substance was found and was used only in analyses of the association between food and substance type. It was classified into 24 categories based on the Korean Society of Nutrition’s classification³⁶ and the Korean Food Promotion Institute’s international Korean food menu guide³⁷, including rice, noodles, soups and stews, kimchi, fried foods, fruits, etc.

Statistical analysis

The overall study design and analytic workflow are summarized (Fig. 1). Specifically, statistical analysis was conducted in the following sequence. First, an exploratory analysis was performed using chi-square tests and a

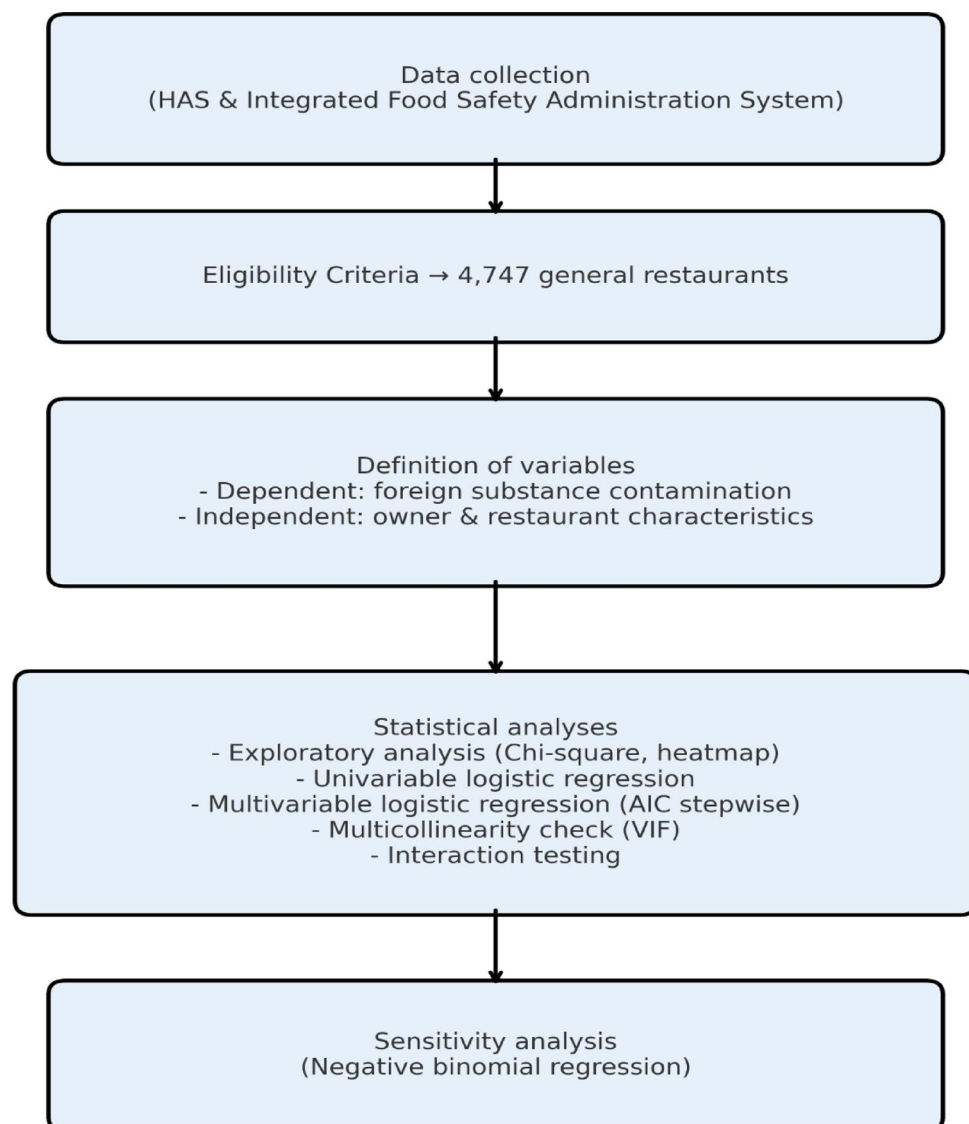


Fig. 1. Flowchart of the study methodology. The chart summarizes the process of data collection, inclusion/exclusion criteria, variable definition, including the dependent variable (foreign substance contamination), statistical analyses (exploratory, univariable and multivariable logistic regression, multicollinearity check, and interaction testing), and sensitivity analysis using negative binomial regression.

heatmap based on standardized residuals to examine the distribution patterns between food categories and types of foreign substances. Univariable logistic regression analyses were then conducted to assess the associations between each independent variable and the presence of foreign substance contamination. To identify key predictors while minimizing overfitting, a multivariable logistic regression model was constructed using stepwise selection based on the Akaike Information Criterion (AIC). The general form of the logistic regression model applied in this study was defined as:

$$\text{logit}(p) = \ln(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

where p denotes the probability of foreign substance contamination. Logistic regression assumptions were also examined, including independence of observations, linearity of continuous predictors in the logit, absence of influential outliers, absence of multicollinearity (all variance inflation factors < 10), and adequacy of sample size. The final model also included interaction terms between key variables, of which one statistically significant interactions were retained. To assess the robustness of the findings, a sensitivity analysis was conducted using a negative binomial regression model, in which the number of contamination reports was used as the dependent variable.

Logistic regression modeling, interaction analysis, and sensitivity analysis were all conducted using SAS 9.4 (SAS Institute Inc., Cary, NC, USA), and chi-square testing and data visualization were performed using Python 3.12.4. All statistical analyses were two-sided, and a p -value of < 0.05 was considered statistically significant.

Results

Among the 4,747 general restaurants analyzed, 343 restaurants (7.2%) had reported incidents of foreign substance contamination in food. As presented in Table 1, restaurants with contamination reports were more likely to have male owners (61.2% vs. 49.0%) and were predominantly operated by individuals in their 30s and 40s. These restaurants also had a higher prevalence of joint ownership by two or more individuals (10.2% vs. 4.1%), were more frequently categorized as serving Chinese cuisine (16.9% vs. 3.1%), and were more commonly located on the first floor. While no notable differences were observed in terms of administrative district or restaurant area, the mean duration of operation was significantly shorter in the contamination group compared to the non-contamination group (4.87 vs. 6.13 years).

Foreign substances were most frequently detected in fried dishes, soups and stews, noodles and dumplings, and stir-fried dishes (Fig. 2A). In terms of types of contaminants, hair was the most commonly reported, followed by insects, metal, plastic, and plastic wrap (Fig. 2B). By food categories, hair was predominantly found in fried dishes, rice dishes, and stir-fried items; insects were most frequently found in soups, noodles and dumplings, and rice dishes; metal in fried dishes and noodles/dumplings; plastic in rice and noodles/dumplings; and plastic wrap in fried dishes, rice, and soups (Table 2). Standardized residual analysis comparing observed versus expected frequencies revealed that mold was overrepresented in fruits, rubber in beverages/alcohol/teas and stews, stones in grilled foods, bones in pan-fried dishes, and glass fragments in vegetable dishes. Conversely, insects were underrepresented in fried foods (Fig. 3).

Univariate logistic regression analysis revealed significant associations between foreign substance contamination and several variables, including gender, age, number of owners, use of delivery platforms, operation duration, cuisine type, and floor level of the restaurant (Table 3). Restaurants operated by men, individuals in their 30s, those with joint ownership, and those using delivery platforms had higher odds of contamination. Shorter operation duration (by owner), Chinese cuisine, and first-floor location were also significantly associated with increased contamination risk.

Multivariate logistic regression analysis identified cuisine type as the most powerful predictor of contamination (Table 4). Restaurants serving Chinese cuisine had a 6.1-fold higher likelihood of contamination compared to those serving Korean cuisine (OR = 6.11, 95% CI = 4.15–9.00). Restaurants using delivery platforms had 3.6 times higher odds than those not using them (OR = 3.61, 95% CI = 2.84–4.58). Establishments operated by individuals in their 30s showed a 3.5-fold higher risk compared to those operated by individuals in their 70s (OR = 3.47, 95% CI = 1.48–8.14), while jointly operated establishments showed a 1.8-fold higher risk than those operated by a single individual (OR = 1.79, 95% CI = 1.18–2.72). All VIF values were below 2, indicating no multicollinearity among the variables. To validate these findings, a negative binomial regression was also conducted using the number of contamination reports as the outcome variable (Table 5), yielding comparable results to those from the logistic regression model.

The final model included one statistically significant interaction term. Potential two-way interaction effects were examined among the candidate variables, and only the interaction between age group and number of owners showed a significant association with the likelihood of foreign substance contamination. For this interaction (Fig. 4), restaurants operated by a single owner exhibited a higher risk of contamination among those in their 20s to 50s, whereas a substantial reduction in risk was observed in jointly operated restaurants within the 40s age group (estimate = -4.22, p = 0.0003). In contrast, no significant difference was found in jointly operated restaurants among those in their 30s, and among owners in their 60s, the number of owners did not significantly affect contamination risk.

Based on the final logistic regression model, the highest predicted probability of foreign substance contamination reports was observed in the combination of restaurant serving Chinese cuisine, owners in their 30s, jointly operated by two or more people, and using food delivery platforms, with a predicted probability of 94.8%. In contrast, the lowest predicted probability was found in the combination of other cuisine types, owners in their 70s, single owner, and no use of food delivery platforms, with a probability of only 0.6%.

According to this final model, the predicted log odds of reporting foreign substance contamination in restaurant food can be estimated as follows:

Characteristic	Contamination status		P value
	Contaminated (N = 343)	Not contaminated (N = 4404)	
Gender			< 0.001
Male	61.2 (210)	49.0 (2159)	
Female	38.8 (133)	51.0 (2245)	
Age group			< 0.001
≤ 29	5.2 (18)	3.4 (150)	
30–39	28.3 (97)	17.8 (783)	
40–49	31.5 (108)	23.9 (1,054)	
50–59	22.2 (76)	27.2 (1200)	
60–69	11.1 (38)	21.6 (950)	
≥ 70	1.7 (6)	6.1 (267)	
Number of owners			< 0.001
One	89.8 (308)	95.9 (4223)	
≥ Two	10.2 (35)	4.1 (181)	
Area (100 m ²)	0.67 (0.6)	0.69 (1.1)	0.82
Use of food delivery platform			< 0.000
Yes	56.1 (193)	25.0 (1100)	
No	43.9 (151)	75.0 (3303)	
Operating duration (years)			
Owner	4.87 (4.48)	6.13 (5.65)	0.005
Location	8.19 (7.33)	9.56 (7.97)	0.44
Type of cuisine			< 0.000
Chinese cuisine	16.9 (58)	3.1 (138)	
Japanese cuisine	4.1 (14)	5.3 (232)	
Korean Bunsik *	6.4 (22)	9.0 (396)	
Korean cuisine	37.6 (129)	41.0 (1807)	
Korean fried chicken	14.0 (48)	15.3 (674)	
Western cuisine	5.2 (18)	5.9 (262)	
Other	15.8 (54)	20.4 (896)	
Restaurant floor location			0.048
Basement	7.6 (26)	11.1 (488)	
1st floor	78.2 (269)	72.8 (3,206)	
≥ 2nd floor	14.2 (49)	15.8 (695)	
Mixed [†]	-	0.3 (14)	
Administrative subdistrict			0.23
A	25.9 (89)	21.2 (935)	
B	42.2 (144)	46.4 (2,045)	
C	14.2 (49)	15.7 (691)	
D	5.8 (20)	4.8 (209)	
E	11.9 (41)	11.9 (524)	

Table 1. Characteristics of restaurants based on foreign substance contamination in food. Values are presented as n (%) or mean (SD). p-values were calculated using the Chi-squared test. *Bunsik refers to a type of Korean eatery that sells light meals such as gimhap, tteokbokki, instant noodles, and other similar dishes. [†]“Mixed” refers to restaurants located across multiple floors (e.g., basement and first floor). This category was excluded from the analysis due to classification ambiguity.

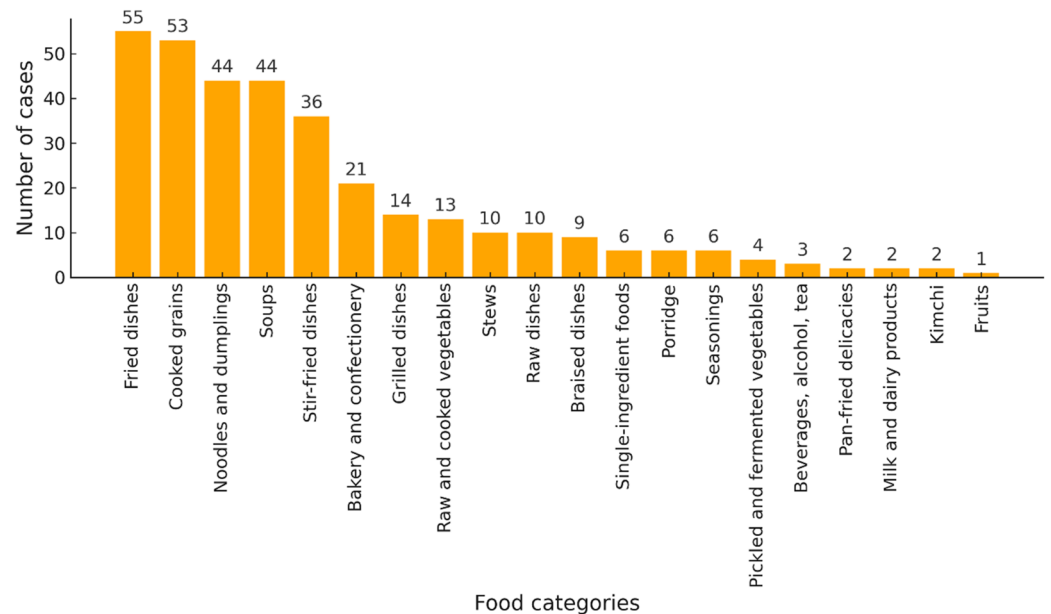
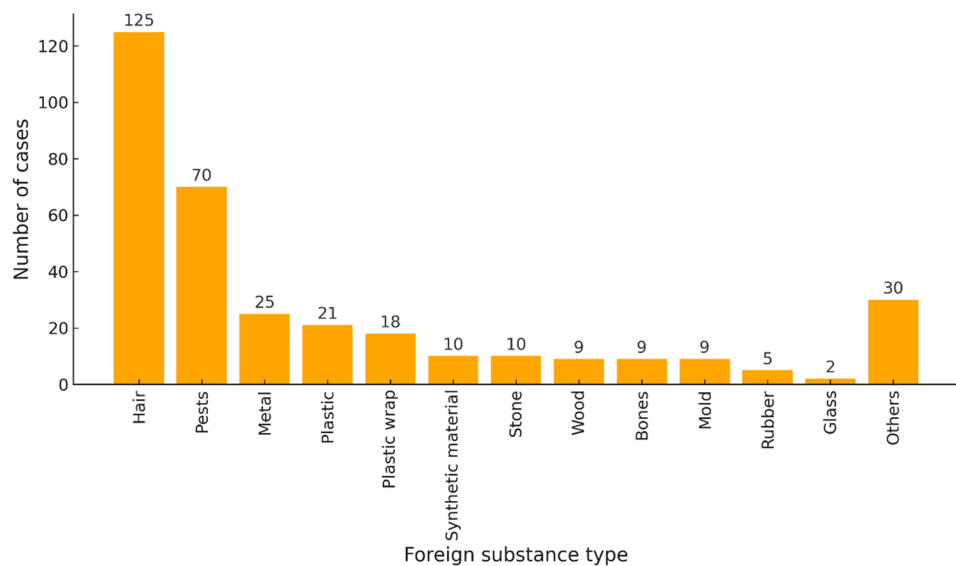
A**B**

Fig. 2. Frequency of foreign substance contamination by (A) food categories and (B) foreign substance types.

$$\begin{aligned}
 \text{logit}(p) = & -4.74 + 1.81 \text{ age_20s} + 1.71 \text{ age_30s} + 1.76 \text{ age_40s} + 1.40 \text{ age_50s} \\
 & + 1.04 \text{ age_60s} + 2.83 \text{ owner_2plus} + 1.30 \text{ del_app_yes} + 1.81 \text{ food_chinese} \\
 & + 0.39 \text{ food_japanese} + 0.22 \text{ food_bunsik} + 0.10 \text{ food_chicken} \\
 & + 0.18 \text{ food_western} - 0.38 \text{ food_other} + 0.39 \text{ floor_1st} - 0.09 \text{ floor_2nd_or_above} \\
 & - 2.49 (\text{age_20s} \times \text{owner_2plus}) - 1.48 (\text{age_30s} \times \text{owner_2plus}) \\
 & - 4.22 (\text{age_40s} \times \text{owner_2plus}) - 3.88 (\text{age_50s} \times \text{owner_2plus}) \\
 & - 1.66 (\text{age_60s} \times \text{owner_2plus})
 \end{aligned}$$

Where 'age_20s' to 'age_60s' are dummy variables coded as 1 if the owner belongs to the corresponding age group (i.e., 20s to 60s), and 0 otherwise (reference group: owners aged 70 or older), 'owner_2plus' is coded as 1 if the restaurant is jointly operated by two or more individuals, and 0 if operated by a single owner, 'del_

	Bones	Glass	Hair	Metal	Mold	Pests	Plastic	Plastic wrap	Rubber	Stone	Synthetic material	Wood	Others	Total
Cooked grains	1 (11.1)	0 (0)	24 (19.2)	1 (4)	1 (11.1)	10 (14.3)	7 (33.3)	3 (16.7)	0 (0)	1 (10)	2 (20)	1 (11.1)	2 (6.7)	53 (15.5)
Stir-fried dishes	0 (0)	1 (50)	16 (12.8)	1 (4)	0 (0)	6 (8.6)	1 (4.8)	1 (5.6)	1 (20)	4 (40)	0 (0)	2 (22.2)	3 (10.5)	36 (10.5)
Bakery and confectionery	1 (11.1)	0 (0)	10 (8)	0 (0)	0 (0)	5 (7.1)	2 (9.5)	0 (0)	0 (0)	0 (0)	0 (0)	1 (11.1)	2 (6.7)	21 (6.7)
Beverages, alcohol, tea	0 (0)	0 (0)	0 (0)	0 (0)	0 (4)	1 (1.4)	0 (0)	1 (5.6)	1 (20)	0 (0)	0 (0)	0 (0)	0 (0)	3 (0.9)
Fruits	0 (0)	0 (0)	0 (0)	0 (0)	1 (11.1)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (0.3)
Grilled dishes	0 (0)	0 (0)	5 (4)	0 (0)	0 (0)	1 (1.4)	1 (4.8)	2 (11.1)	0 (0)	3 (30)	1 (10)	0 (0)	1 (3.3)	14 (4.1)
Soups	1 (11.1)	0 (0)	15 (12)	3 (12)	1 (11.1)	12 (17.1)	3 (14.3)	3 (16.7)	0 (0)	1 (10)	0 (0)	0 (0)	5 (16.7)	44 (12.8)
Raw dishes	0 (0)	0 (0)	3 (2.4)	2 (8)	0 (0)	3 (4.3)	1 (4.8)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (3.3)	10 (2.9)
Pickled and fermented vegetables	0 (0)	0 (0)	2 (1.6)	0 (0)	1 (11.1)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (3.3)	4 (1.2)
Pan-fried delicacies	1 (11.1)	0 (0)	1 (0.8)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2 (0.6)
Stews	0 (0)	0 (0)	2 (1.6)	2 (8)	0 (0)	2 (2.9)	0 (0)	1 (5.6)	2 (40)	0 (0)	1 (10)	0 (0)	3 (10)	13 (3.8)
Braised dishes	0 (0)	0 (0)	4 (3.2)	1 (4)	0 (0)	2 (2.9)	0 (0)	0 (0)	1 (10)	0 (0)	0 (0)	1 (11.1)	0 (0)	9 (2.6)
Glazed dishes	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (1.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (3.3)	2 (0.6)
Porridge	1 (11.1)	0 (0)	1 (0.8)	0 (0)	1 (11.1)	2 (2.9)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (3.3)	6 (1.7)
Kimchi	0 (0)	0 (0)	1 (0.8)	0 (0)	0 (0)	1 (1.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2 (0.6)
Noodles and dumplings	2 (22.2)	0 (0)	10 (8)	4 (16)	1 (11.1)	11 (15.7)	4 (19)	1 (5.6)	0 (0)	0 (0)	3 (30)	2 (22.2)	6 (20)	44 (12.8)
Milk and dairy products	0 (0)	0 (0)	2 (1.6)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2 (0.6)
Raw and cooked vegetables	0 (0)	1 (50)	2 (1.6)	0 (0)	0 (0)	4 (5.7)	0 (0)	1 (5.6)	0 (0)	1 (10)	0 (0)	0 (0)	1 (3.3)	10 (2.9)
Seasonings	0 (0)	0 (0)	2 (1.6)	1 (4)	1 (11.1)	2 (2.9)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	6 (1.7)
Single-ingredient foods*	0 (0)	0 (0)	0 (0)	0 (0)	1 (11.1)	5 (7.1)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	6 (1.7)
Fried dishes	2 (22.2)	0 (0)	25 (20)	10 (40)	1 (11.1)	2 (2.9)	2 (9.5)	5 (27.8)	0 (0)	0 (0)	3 (30)	2 (22.2)	3 (10)	55 (16.0)
Total	9 (100)	2 (100)	125 (100)	25 (100)	9 (100)	70 (100)	21 (100)	18 (100)	5 (100)	10 (100)	10 (100)	9 (100)	30 (100)	343 (100)

Table 2. Presence of foreign substances in food categories. () indicates the proportion of the foreign substance in each food category. *Single-ingredient foods refer to foods consisting of a single raw material without additional seasonings or a combination of multiple ingredients, such as fresh vegetables, nuts, chicken breast, and roasted sweet potatoes.

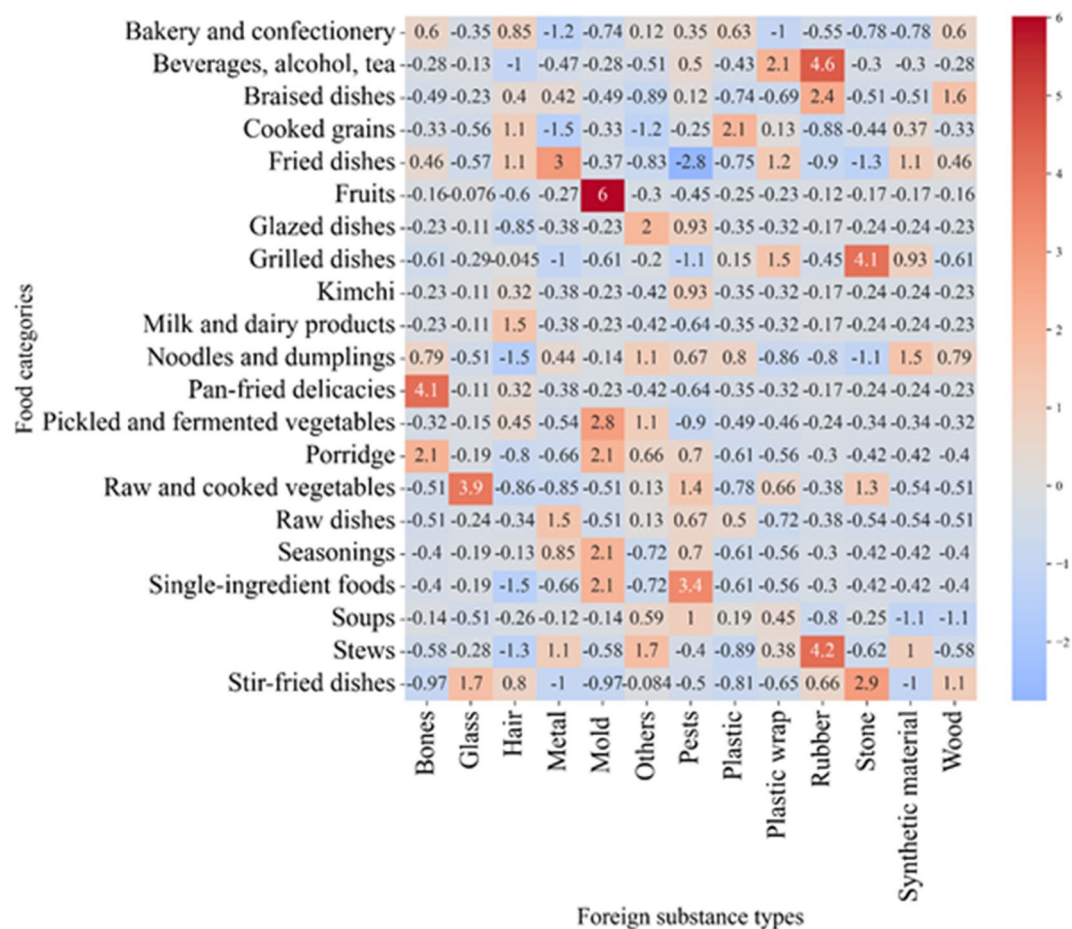


Fig. 3. Heatmap of standardized residuals representing the residuals between food categories and foreign substance types. Red cells indicate positive residuals, showing that the frequency between food categories and foreign substance types is higher than expected. Blue cells indicate negative residuals, showing that the frequency between food categories and foreign substance types is lower than expected. Single-ingredient foods refer to foods consisting of a single raw material without additional seasonings or a combination of multiple ingredients, such as fresh vegetables, nuts, chicken breast, and roasted sweet potatoes.

app_yes' is 1 if the restaurant is registered on a third-party food delivery platform such as Baemin, Yogiyo, or Coupang Eats, and 0 otherwise, 'food_chinese' to 'food_other' are coded as 1 if the restaurant primarily serves the corresponding cuisine, and 0 otherwise (reference: Korean cuisine), 'floor_1st' and 'floor_2nd_or_above' are coded as 1 if the restaurant is located on the respective floor, and 0 otherwise (reference: basement level), and interaction terms (e.g., 'age_30s × owner_2plus') represent the combined effects of age group and ownership structure beyond their individual contributions.

Discussion

An analysis of risk factors for foreign substance contamination in urban dining environments revealed that restaurants serving Chinese cuisine, using food delivery platforms, operated by owners in their 30s, or managed jointly exhibited significantly higher rates of contamination reports. In addition to these findings, the following discussion addresses the study's limitations as well as the broader social and methodological implications.

Cuisine type (Chinese cuisine)

Restaurants serving Chinese cuisine exhibited the highest reporting rates of foreign substance contamination, which is consistent with previous studies. Logistic regression-based predictions also showed that all of the top five high-risk combinations involved restaurants serving Chinese food. While earlier research has reported relatively lower hygiene scores in Chinese restaurants³³, this study did not include variables such as owner nationality or ethnicity. In this context, the term 'Chinese cuisine' refers solely to the food category, not to cultural or ethnic identity. Therefore, this observed association should not be interpreted as a direct causal relationship. Instead, cuisine type may serve as a proxy indicator for certain cooking environments or operational patterns.

For instance, Chinese cuisine typically involves high-temperature cooking methods such as frying or stir-frying, rapid preparation, and complex recipes, all of which may increase the risk of physical hazards such as equipment wear, splattering, or handling errors. In addition, Chinese restaurants in Korea often offer a wide variety

Variable	Comparison category	OR	95% CI	P value (vs. ref)	P value
Gender					
Female	Male	0.61	0.48–0.76	<0.001	<0.001
Age group					
≤29	≥70	5.34	2.07–13.74	0.001	
30–39		5.51	2.39–12.72	<0.001	
40–49		4.61	2.00–10.60	0.001	
50–59		2.82	1.21–6.54	0.016	
60–69		1.78	0.74–4.26	0.019	
Number of owners					
≥ Two	One	2.64	1.81–3.86	<0.001	<0.001
Area (100m ²)		0.99	0.88–1.10	0.8	0.8
Use of third-party food delivery platform					
Yes	No	3.84	3.07–4.80	<0.001	<0.001
Operating duration (10Years)					
Owner		0.62	0.49–0.78	<0.001	<0.001
Location		0.8	0.69–0.92	0.003	0.003
Type of cuisine					
Chinese cuisine	Korean cuisine	5.89	4.13–8.40	<0.001	<0.001
Japanese cuisine		0.85	0.48–1.50	0.56	
Korean Bunsik*		0.78	0.49–1.24	0.29	
Korean fried chicken		0.99	0.71–1.41	0.99	
Western cuisine		0.97	0.58–1.61	0.89	
Other		0.86	0.62–1.20	0.36	
Restaurant floor location					
1st floor	Basement	1.58	1.04–2.38	0.031	0.048
≥2nd floor		1.32	0.81–2.16	0.26	
Administrative subdistrict					
B	A	0.75	0.57–0.98	0.036	0.23
C		0.74	0.52–1.07	0.11	
D		1.01	0.61–1.67	0.98	
E		0.82	0.56–1.21	0.32	

Table 3. Univariable logistic regression model for foreign substance contamination. *Korean Bunsik refers to a type of Korean eatery that sells light meals such as gimhap, tteokbokki, instant noodles, and other similar dishes.

of dishes at relatively low prices, which may contribute to time pressure and workload, potentially compromising hygiene management³⁸. Some of these restaurants may also rely heavily on food delivery platforms and operate as dark kitchens with limited sanitation infrastructure³⁹. As such, the observed association is more likely to stem from the cooking and operational characteristics commonly found in Chinese restaurants, rather than the cuisine itself. Future research should include contextual variables—such as menu pricing, recipe complexity, kitchen environment, operator experience, and hygiene inspection outcomes—to more precisely investigate this relationship.

Use of food delivery platforms

The higher contamination risk in restaurants using delivery platforms is likely linked to structural transformations in the food service industry. This may reflect not only vulnerabilities in the delivery-based cooking environment but also reporting bias due to more active complaint behavior among digital consumers. Customers who order through apps may be more inclined to report hygiene issues via digital channels, increasing external scrutiny compared to traditional dine-in establishments.

The rapid emergence of “dark kitchens” or “ghost kitchens” following the COVID-19 pandemic has led to the proliferation of establishments that only provide cooking and packaging services without dine-in areas^{8–12}. Although these businesses are often indistinguishable from traditional restaurants to consumers, they possess distinct hygiene vulnerabilities. Because the food is prepared in spaces that consumers cannot observe, and often in constrained environments with inadequate ventilation, hot water, or sanitation equipment^{13–17}, the risk of contamination is inherently higher. Additionally, the handling practices of third-party delivery personnel may introduce further risk of secondary contamination during transport¹⁷. Consumers have also voiced concern over the hygiene of dark kitchens^{18,19}. From a policy perspective, these findings highlight the importance of transparent disclosure systems that can bridge the information gap between consumers and non-visible kitchen environments. For example, the United Kingdom’s Food Hygiene Rating Scheme⁴⁰, which publicly discloses

Variable	Comparison category	OR	95% CI	P value (vs. ref)	P value
Age group					
≤ 29	≥ 70	3.35	1.27–8.84	0.015	< 0.001
30–39		3.47	1.48–8.14	0.004	
40–49		2.97	1.27–6.92	0.012	
50–59		2.12	0.90–4.98	0.08	
60–69		1.62	0.67–3.91	0.28	
Number of owners					
≥ Two	One	1.79	1.18–2.72	0.006	0.006
Use of third-party food delivery platform					
Yes	No	3.61	2.84–4.58	< 0.001	< 0.001
Type of cuisine					
Chinese cuisine	Korean cuisine	6.11	4.15–9.00	< 0.001	< 0.001
Japanese cuisine		0.6	0.33–1.07	0.08	
Korean Bunsik*		0.68	0.43–1.10	0.11	
Korean fried chicken		0.9	0.64–1.29	0.57	
Western cuisine		0.76	0.45–1.29	0.31	
Other		0.68	0.49–0.96	0.027	
Restaurant floor location					
1st Floor	Basement	1.48	0.96–2.28	0.08	0.008
		0.91	0.54–1.53	0.71	

Table 4. Multivariable logistic regression model for foreign substance contamination. *Korean Bunsik refers to a type of Korean eatery that sells light meals such as gimbap, tteokbokki, instant noodles, and other similar dishes.

Variable	Comparison category	IRR	95% CI	pvalue
Age group				
≤ 29	≥ 70	3.74	1.31–10.62	0.013
30–39		4.99	1.96–12.71	< 0.0001
40–49		3.77	1.70–8.94	0.003
50–59		3.27	1.39–7.69	0.007
60–69		1.77	0.80–3.90	0.16
Number of owners				
≥ Two	One	2.34	1.42–3.83	< 0.0001
Use of third-party food delivery platform				
Yes	No	6.40	5.00–8.35	< 0.0001
Type of Cuisine				
Chinese Cuisine	Korean Cuisine	9.28	5.92–14.56	< 0.0001
Japanese Cuisine		0.56	0.30–1.03	0.07
Korean Bunsik*		0.84	0.52–1.36	0.47
Korean Fried Chicken		0.87	0.60–1.26	0.46
Western Cuisine		0.61	0.34–1.09	0.10
Other		0.62	0.44–0.88	0.008
Restaurant floor location				
1st floor	Basement	0.65	0.48–0.88	0.005
≥ 2nd floor		0.86	0.50–1.50	0.62

Table 5. Multivariable negative binomial regression model for foreign substance contamination. *Korean Bunsik refers to a type of Korean eatery that sells light meals such as gimbap, tteokbokki, instant noodles, and other similar dishes. This table presents incidence rate ratios (IRRs) from a multivariable negative binomial regression model predicting the number of foreign substance contamination reports based on restaurant and owner characteristics. Estimates were adjusted for all covariates included in the model. CI = confidence interval.

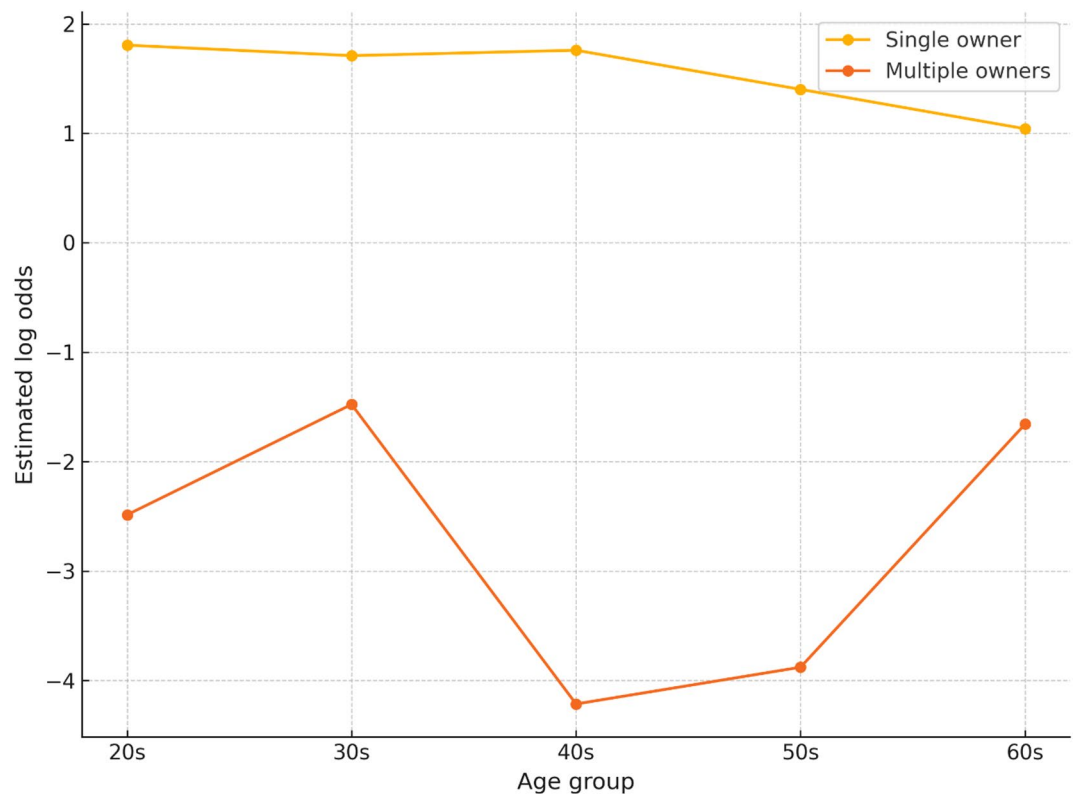


Fig. 4. Interaction effect between age group and number of owners on the log odds of foreign substance contamination. Joint ownership showed significantly lower risk in the 40s age group, while no significant differences were observed in the 30s and 60s.

hygiene ratings and applies differentiated evaluation standards, has been shown to effectively encourage voluntary compliance among operators while enhancing consumer trust. Future research should further clarify this association by comparing pre- and post-pandemic data and incorporating variables such as the presence of customer-facing spaces and the proportion of sales generated through delivery services.

Owner age

The finding that restaurants operated by individuals in their 30s had the highest rate of contamination reports is noteworthy. Previous studies indicate that older food handlers tend to have greater food safety knowledge and better hygiene practices^{41,42}, and that food safety knowledge positively affects behavior⁴³. Significant positive associations between food safety attitudes and knowledge have also been reported^{44,45}. Cognitive and psychological differences by age may also influence hygiene behavior⁴⁶. Younger adults (20–30 s) are more likely to exhibit overconfidence, sensation-seeking, and prioritize short-term efficiency over long-term outcomes^{47,48}. These traits may reduce the likelihood that hygiene education translates into consistent practice. In light of this, future research should incorporate hygiene-related training history, field experience, and psychological traits for a more integrated analysis.

Joint ownership structures

Restaurants jointly operated by two or more individuals exhibited higher rates of contamination reports. This finding suggests that ambiguity in hygiene responsibility may lead to a decline in overall hygiene standards. Prior studies have noted that multi-owner establishments often suffer from inconsistent enforcement of hygiene rules, unclear accountability, interpersonal conflicts, and weakened system management^{49,50}. However, the interaction analysis in this study revealed that such risks varied by context. For example, restaurants jointly operated by individuals in their 40s showed the greatest reduction in contamination risk, whereas no significant difference was observed between single and jointly operated establishments among those in their 30s. These results indicate that the impact of joint operation on hygiene risk may either increase or mitigate the risk depending on contextual factors such as the age of the restaurant owner.

Accordingly, jointly operated restaurants may benefit from more structured hygiene management systems, including clearly defined hygiene responsibilities, the appointment of a dedicated hygiene supervisor, and regular internal inspections. For instance, the Singapore Food Agency (SFA) mandates the designation of a qualified Food Hygiene Officer (FHO)-certified at WSQ Level 3 or higher-for high-risk or high-traffic establishments, offering a practical policy model for enhancing hygiene supervision in businesses with complex ownership structures⁴⁸. Although restaurant floor area was not significantly associated with the risk of foreign

substance contamination, this finding suggests a need to reconsider the common belief that larger kitchens are inherently more hygienic. Instead, our results suggest that factors such as food pricing, menu complexity, kitchen environment, and managerial experience may play a more critical role in maintaining food safety^{52–55}. Future studies should further investigate how these diverse factors influence hygiene conditions in restaurant settings.

Strengths and limitations

The primary strength of this study lies in its use of administrative data covering the entire restaurant population, rather than individual case reports, to empirically examine complex and context-specific risk factors. This approach enabled a comparison between restaurants with and without contamination reports, offering practical evidence to inform food safety policy.

However, several limitations should be noted. First, the analysis was geographically limited to a single district in Seoul, which limits the generalizability of the findings. Second, our study did not distinguish between different types of food preparation environments, such as delivery-only and mixed-operation restaurants. Third, as the analysis relied on contamination reports submitted to administrative systems, reporting bias may have been present. Fourth, the absence of hygiene inspection data made it difficult to triangulate the findings with actual sanitation conditions. Fifth, the lack of data on hygiene training or food handler behavior limited the ability to explore the mechanisms of contamination in greater depth.

Conclusion

This study analyzed 4,747 general restaurants located in a district of Seoul and identified four key factors significantly associated with reports of foreign substance contamination: serving Chinese cuisine, use of food delivery platforms, restaurant owners in their 30s, and joint ownership structures. These findings suggest that contamination in restaurant-prepared food does not occur randomly, but is influenced by complex and context-dependent conditions.

Future research should expand the scope of analysis to include multiple regions and time periods, and consider comparisons before and after the COVID-19 pandemic to evaluate the effects of delivery platform expansion on food safety. Additionally, machine learning models incorporating a wide range of administrative and contextual variables—such as hygiene inspection results, menu complexity, food pricing, kitchen environment, and operator experience—could be useful for precisely predicting contamination risk and designing preventive strategies.

From a policy perspective, implementing standardized hygiene training systems applicable to both traditional and delivery-based restaurants may offer a more feasible and sustainable approach to food safety management.

Data availability

The data that support the findings of this study are available from the Integrated Food Safety Information Network of the Ministry of Food and Drug Safety (<https://www.mfds.go.kr/eng/index.do>) but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Jiwon Lee (lgw2611@nowon.go.kr).

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Author contributions

Myungsook Jung contributed to data collection, study design and planning, data analysis, and drafting both the initial and main content of the manuscript. Sangshin Park contributed to study design and planning, supervised the data analysis, and revised the main content of the manuscript. All authors reviewed and approved the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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